

ARYAN SAIGAL

AI/ML Engineer · Full-Stack Developer · Cloud & DevOps
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PROFESSIONAL SUMMARY

AI/ML engineer and solutions architect who designs and ships production AI systems end to end, from RAG pipelines and LLM agent integrations to the full-stack products built around them. Experienced delivering document intelligence platforms, multilingual retrieval systems, real-time voice AI and Text-to-SQL interfaces using LangChain, FastAPI and Next.js, extended through MCP servers for multi-modal generation. Owns the full delivery path from architecture and data pipelines through to containerized deployment on GCP Cloud Run with GitHub Actions CI/CD. B.Tech in Computer Science and AWS Certified Cloud Practitioner.

TECHNICAL SKILLS

AI / ML LangChain, RAG pipelines, ChromaDB, OpenRouter, MCP servers, LlamaParse, Docling, spaCy, NLTK, XGBoost, MLflow, generative AI, prompt and agent design

Backend Python, FastAPI, Node.js, REST APIs, SQL, PostgreSQL, SQLite, Supabase

Frontend Next.js 14, React 18, JavaScript, HTML5, CSS3, Tailwind CSS

Cloud & DevOps GCP Cloud Run, AWS, Docker, GitHub Actions CI/CD, Vercel, Cloudflare Workers

Data & Analytics Pandas, NumPy, Matplotlib, Seaborn, Power BI, Excel

Security & Networking Cybersecurity fundamentals, IAM policy review, TCP/IP, IPv4, routing

EXPERIENCE

Solutions Architect Intern · EMB Global (Incuspaze)

May 2026 – Present

Gurgaon, India

- Designed and built AI/ML-powered tools across four project verticals (chatbots, document intelligence, interview automation and SQL interfaces) using LangChain, FastAPI, Next.js and GCP Cloud Run.
- Integrated MCP (Model Context Protocol) servers for multi-modal AI generation and deployed production services via Docker, GitHub Actions CI/CD and GCP Cloud Run.
- Completed 8 Anthropic AI courses in parallel with project delivery, including Building with the Claude API, AI Fluency Framework and Model Context Protocol: Advanced Topics.

ML Intern · Innovate

Aug 2025 – Oct 2025

Remote

- Developed a machine learning model for road traffic management, titled "Predictive Modeling for Road Traffic Management: A Data-Driven Approach", targeting congestion reduction through predictive analytics.
- Built an end-to-end ML pipeline covering data collection, cleaning, feature engineering, model training (regression and classification), hyperparameter tuning and performance evaluation.
- Visualized model outputs and traffic patterns using Matplotlib and Seaborn, producing charts and reports that communicated insights to non-technical stakeholders.

Cloud Infrastructure Intern · CloudSphere

May 2025 – Jul 2025

Noida, India (Hybrid)

- Audited and documented cloud resources and dependencies across multi-cloud environments (AWS, GCP), mapping asset inventories and inter-service relationships.
- Analyzed compute, storage and network usage patterns to identify cost-saving opportunities, preparing optimization reports that highlighted underutilized resources.
- Assisted senior engineers in security assessments, identifying misconfigurations, IAM policy gaps and exposure risks, and configured performance monitoring alerts.

Social Media & Content Intern · The Cocoa Counter

May 2024 – Jun 2024

New Delhi, India

- Managed brand social media accounts across platforms, developed monthly content calendars, wrote captions and collaborated with designers on visuals aligned to brand identity.
- Monitored platform engagement metrics, responded to customer inquiries and implemented feedback-driven changes that improved audience response rates.

PROJECTS

ConversAge AI: RAG Marketing & Content Suite

LangChain, FastAPI, Next.js, ChromaDB, MCP, OpenRouter, Vercel, Cloudflare

- Architected a full-stack RAG pipeline using LangChain and ChromaDB: documents (.pdf, .doc, .ppt) are chunked, embedded and stored as vectors, and user queries retrieve the most relevant context before being passed to the LLM via OpenRouter.
- Extended the chatbot with five capabilities via MCP server integration: AI blog generation, image creation, video generation, a marketing campaign generator and live web search, all accessible from a single Next.js chat interface.
- Secured the application with JWT-based authentication and session management, deploying the frontend on Vercel and the backend on Cloudflare Workers with end-to-end testing across all file formats and generation pipelines.

OCR Multi-Model Parser Platform

FastAPI, LlamaParse, Docling, LiteParser, GCP Cloud Run, Docker, GitHub Actions

- Built a document intelligence platform supporting four OCR engines (LlamaParse, Docling, LiteParser, Document AI) selectable per request, enabling comparison of extraction quality across document types and layouts.
- Added an extract-page route for targeted page-level parsing, an admin dashboard for model configuration and usage monitoring, and full settings and documentation pages covering every parser's parameters.
- Containerized with Docker, deployed to GCP Cloud Run for auto-scaling and set up GitHub Actions CI/CD for automated build and release; also published a standalone CLI tool for running OCR locally.

AI Web Interview Platform

Next.js, FastAPI, OpenAI Realtime Voice API, WebRTC

- Built a browser-based AI interviewer using OpenAI's Realtime Voice API: the system conducts spoken interviews, listens to candidate responses in real time and generates follow-up questions dynamically based on answers.
- Integrated WebRTC for screen sharing and implemented fullscreen enforcement and webcam-based activity analysis, providing proctoring signals to detect suspicious behavior during sessions.
- Developed an admin panel for creating and configuring interview sessions, including question sets, time limits and candidate access controls, with session recordings and activity logs stored for post-interview review.

KANOON: AI-Powered Legal Decision Support System

Python, FastAPI, ChromaDB, React 18, spaCy, NLTK, multilingual NLP (11 Indian languages)

- Designed a RAG pipeline over a curated Indian legal corpus covering IPC sections, FIR procedures and cyber laws, with chunks embedded and retrieved from ChromaDB to ground LLM responses in actual legal text and reduce hallucination.
- Implemented multilingual NLP support across 11 Indian languages using spaCy and NLTK for tokenization, transliteration and language detection, making legal information accessible to non-English-speaking users.
- Delivered a full-stack system (React 18 frontend, FastAPI backend) with a CLI tool for dataset ingestion, and authored an accompanying research paper on AI/ML in Indian legal technology.

Text-to-SQL Chatbot

FastAPI, LangChain, SQLite/PostgreSQL, Next.js

- Built a natural-language-to-SQL interface where users ask questions in plain English, the LLM generates the corresponding SQL query, and the query is validated and executed on a live database backend.
- Implemented a result-formatting layer that converts raw query output (rows, counts, aggregations) into plain-language summaries, and injected table structure into the prompt context to handle multi-step queries, ambiguous phrasing and schema awareness while reducing invalid SQL.

Customer Churn MLOps Pipeline

Python, XGBoost, MLflow, FastAPI, Docker, GitHub Actions, AWS

- Built an end-to-end MLOps pipeline for customer churn prediction using XGBoost, covering data preprocessing, feature engineering, model training, hyperparameter tuning and evaluation.
- Integrated MLflow for experiment tracking and model registry, exposed predictions via a containerized FastAPI inference endpoint, and automated retraining and deployment with GitHub Actions on AWS infrastructure.

Local AI Coding Assistant (Ollama + Qwen Coder)

Ollama, Qwen2.5-Coder, Node.js, Supabase, JavaScript

- Built a local AI coding chatbot running entirely on-device using Ollama and the Qwen2.5-Coder model, with no external API calls, full privacy and zero inference cost.
- Developed a Node.js backend with streaming responses and a browser-based UI, integrating Supabase authentication and persistent chat history across sessions.

EDUCATION

B.Tech, Computer Science Engineering · Amity University, Noida

2022 – 2026

GPA: 7.2 / 10

Class XII, CBSE — 91% (2022) | **Class X, CBSE** — 85% (2020)

Awards: Academic Excellence Award, Grades 9 to 12 (awarded in each academic year).

CERTIFICATIONS

- **AWS Certified Cloud Practitioner** · Amazon Web Services
- **Introduction to Cybersecurity** · Cisco Networking Academy
- **Python Essentials 1 & 2** · Cisco Networking Academy
- **Building with the Claude API** · Anthropic
- **Model Context Protocol: Advanced Topics** · Anthropic
- **Introduction to Model Context Protocol** · Anthropic
- **AI Fluency: Framework & Foundations** · Anthropic
- **Introduction to Agent Skills** · Anthropic
- **AI Capabilities and Limitations** · Anthropic

LANGUAGES

English (full professional) · Hindi (native) · French (read / write)